Open-Source Development

ITIC2262

Presented by:
Lieutenant Commander (IT) RMJP Nayanajith

OPEN OS

The GNU Project and Free Software

* Founded in 1983 by Richard M. Stallman. (RMS)
* Name stands for "GNU's Not Unix!" (a recursive acronym).
* “GNU” is pronounced as “*g’ noo*”
* Goal was Create a completely free, Unix-like operating system



Key Milestones & Components

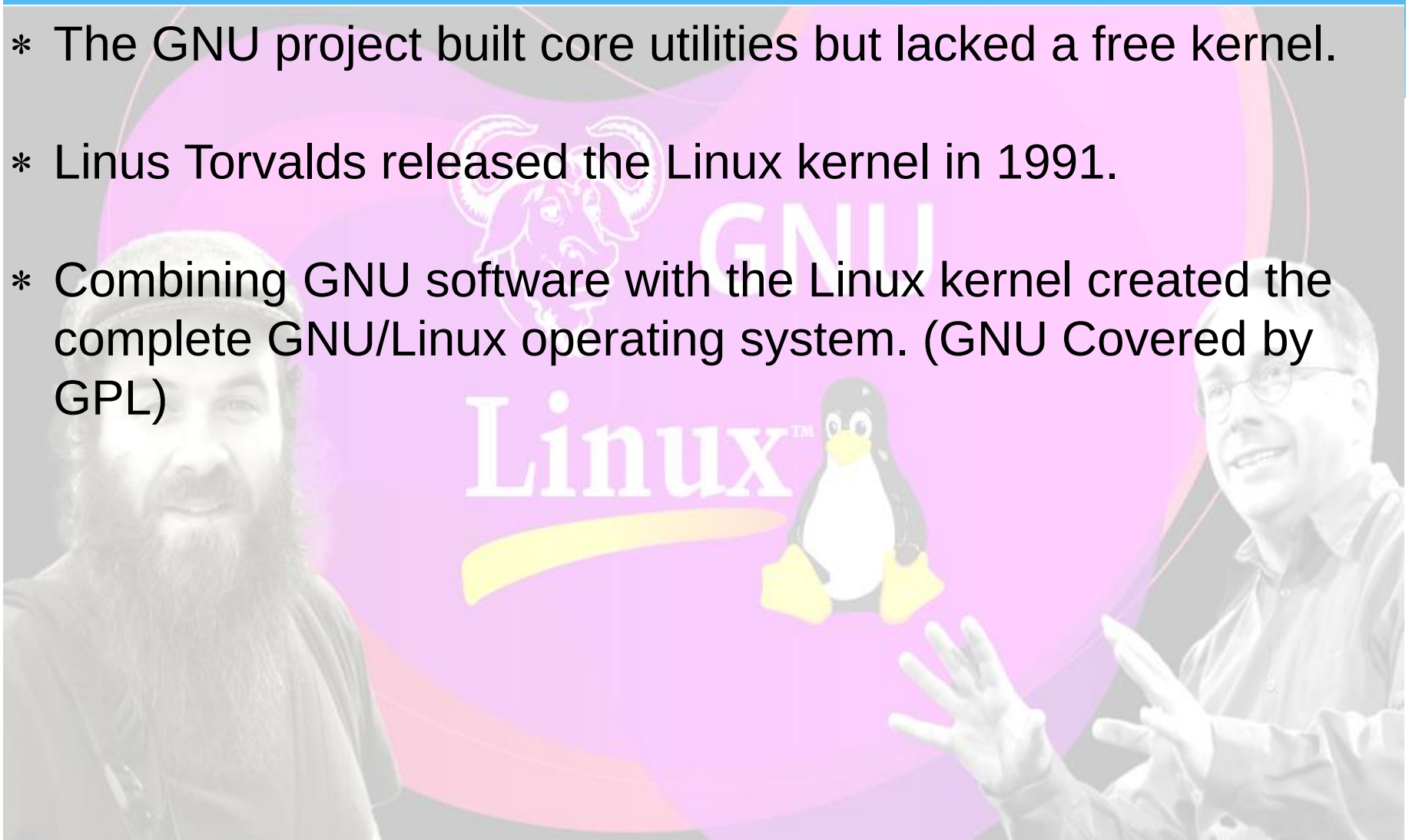
* **1985:** Establishment of the Free Software Foundation (FSF) by Richard Stallman.

* **1989** **:** Introduction of the GNU General Public License (GPL).

* Core Software:
  - GNU Emacs
  - GCC (Compiler)
  - GDB (Debugger) and Bash shell.

The Marriage of GNU and Linux

* The GNU project built core utilities but lacked a free kernel.
* Linus Torvalds released the Linux kernel in 1991.
* Combining GNU software with the Linux kernel created the complete GNU/Linux operating system. (GNU Covered by GPL)



The Four Essential Freedoms

* Freedom 0: Freedom to **Run** the program for any purpose.
* Freedom 1: Freedom to **Study** how the program works and adapt/modify it.
* Freedom 2: Freedom to **Redistribute** copies to help others.
* Freedom 3: Freedom to **Distribute modified** / **Improve and Release** versions to the community

UNIX Principles

* Everything is a File including Hardware.
* All Configuring data store as text.
* Small and Single purpose program.
* Ability to chain programs together to perform complex task.

Linux Origin

* The Linux Operating system was first coded by Finnish computer programmer called Linus Benedict Torvalds in 1991.

* **When he was just 21 !.**



GNU Linux vs BSD





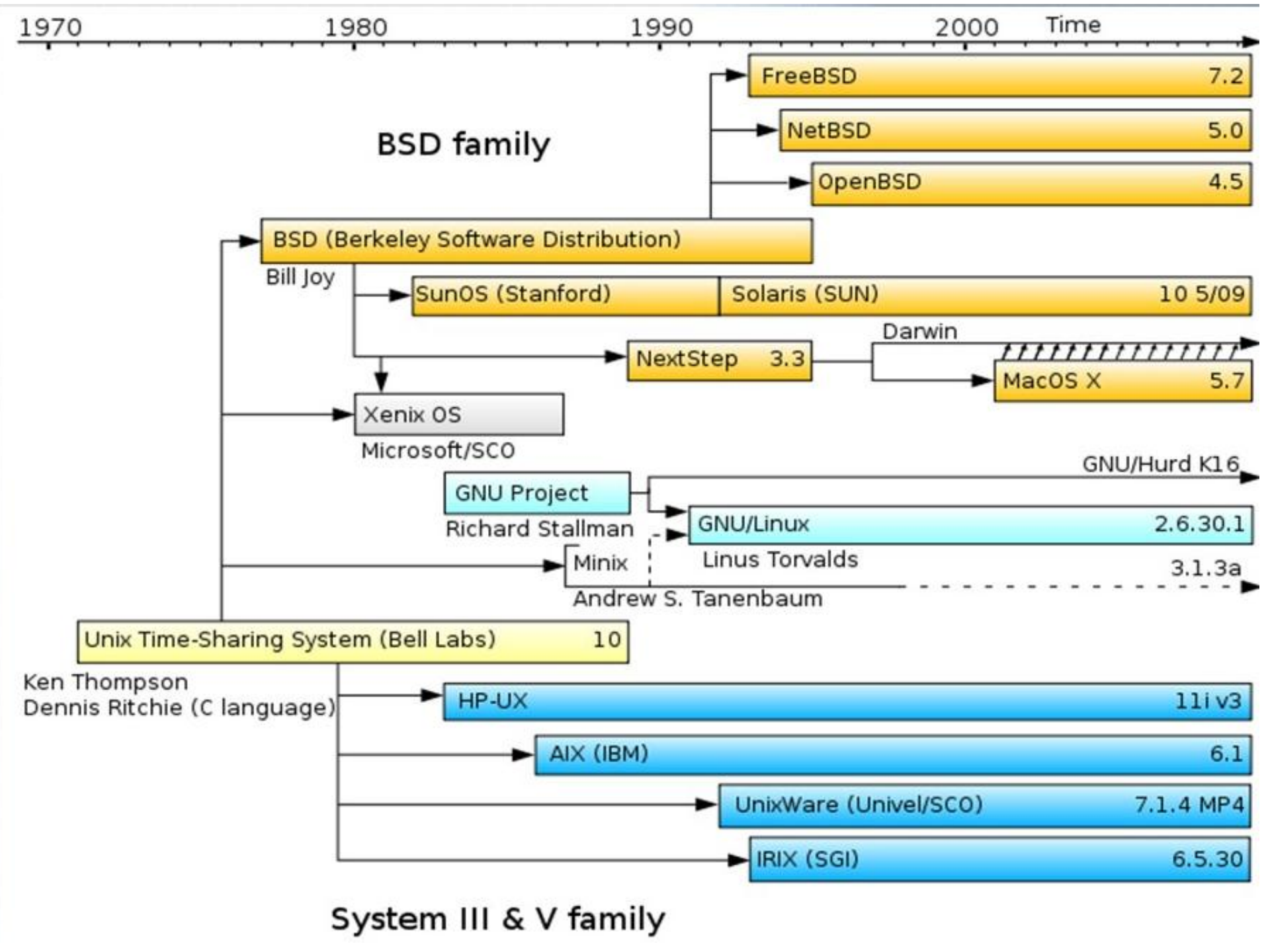
* FreeBSD and GNU Linux both are follow the UNIX philosophy but some differences do exist between the operating systems.

* Linux and FreeBSD use names for hardware files. Linux uses predefined names for each device type, But BSD uses customized names for each device type.

* Free BSD *(**Berkeley Software Distribution**)* Itself modification of *AT&T's UNIX*, and was created by the **University of California**.

* All the pieces of GNU/Linux are developed separately and brought together in distributions.

* FreeBSD has been developed as a complete operating system. *(kernel, device drivers, sysadmin's tool).*


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Time
FreeBSD 7.2
NetBSD 5.0
OpenBSD 4.5
BSD family
BSD (Berkeley Software Distribution)
Bill Joy
SunOS (Stanford)
Solaris (SUN) 10 5/09
Darwin
NextStep 3.3
MacOS X 5.7
Xenix OS
Microsoft/SCO
GNU/Hurd K16
GNU Project
Richard Stallman
GNU/Linux 2.6.30.1
Linus Torvalds
Minix 3.1.3a
Andrew S. Tanenbaum
Unix Time-Sharing System (Bell Labs) 10
Ken Thompson
Dennis Ritchie (C language)
HP-UX 11i v3
AIX (IBM) 6.1
UnixWare (Univel/SCO) 7.1.4 MP4
IRIX (SGI) 6.5.30
System III & V family

The main GNU/Linux dist.

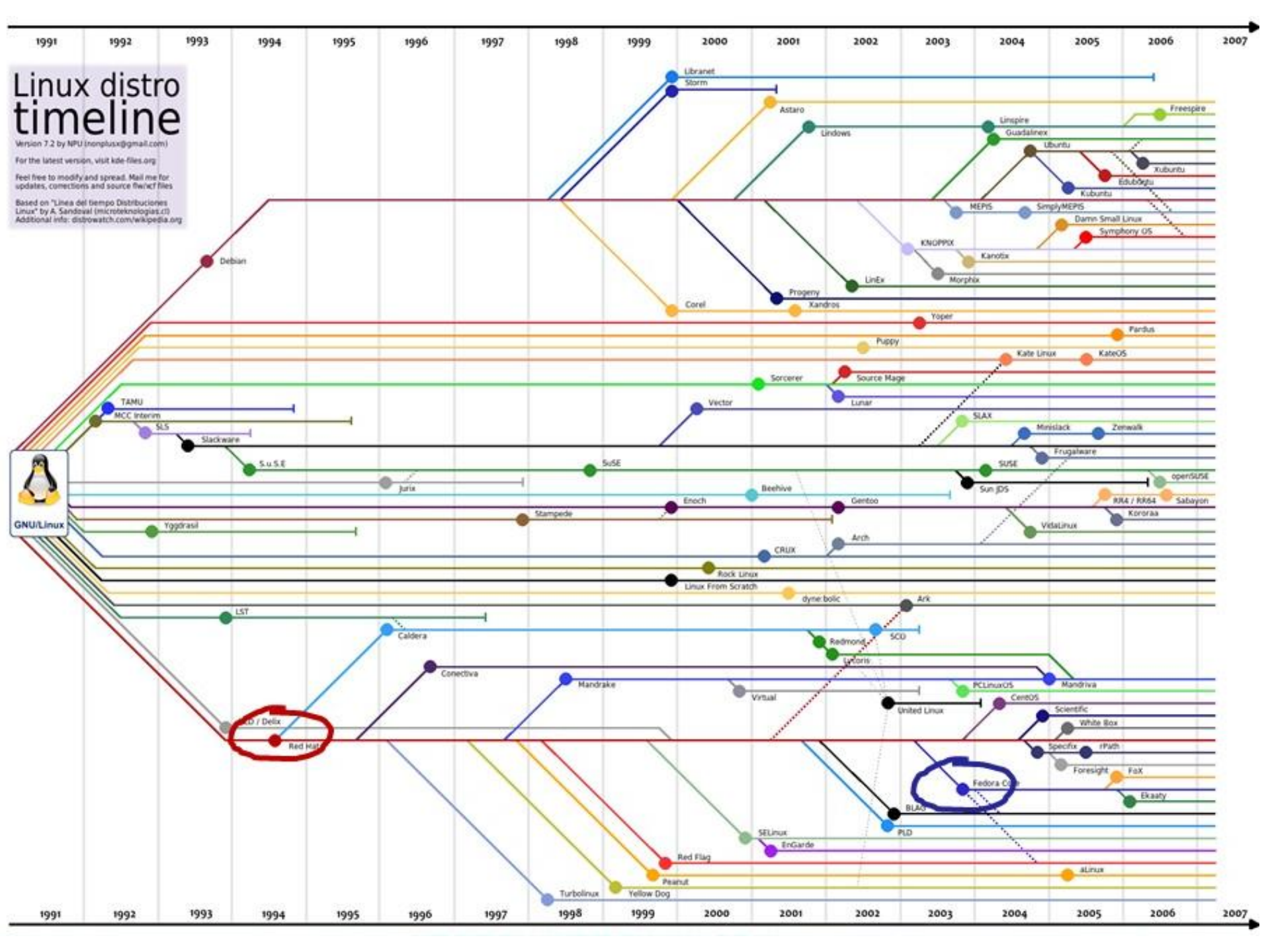
- All have different package management systems
- The differences in performance are negligible. The differences are in management style, how and where files are stored, kernel optimizations.

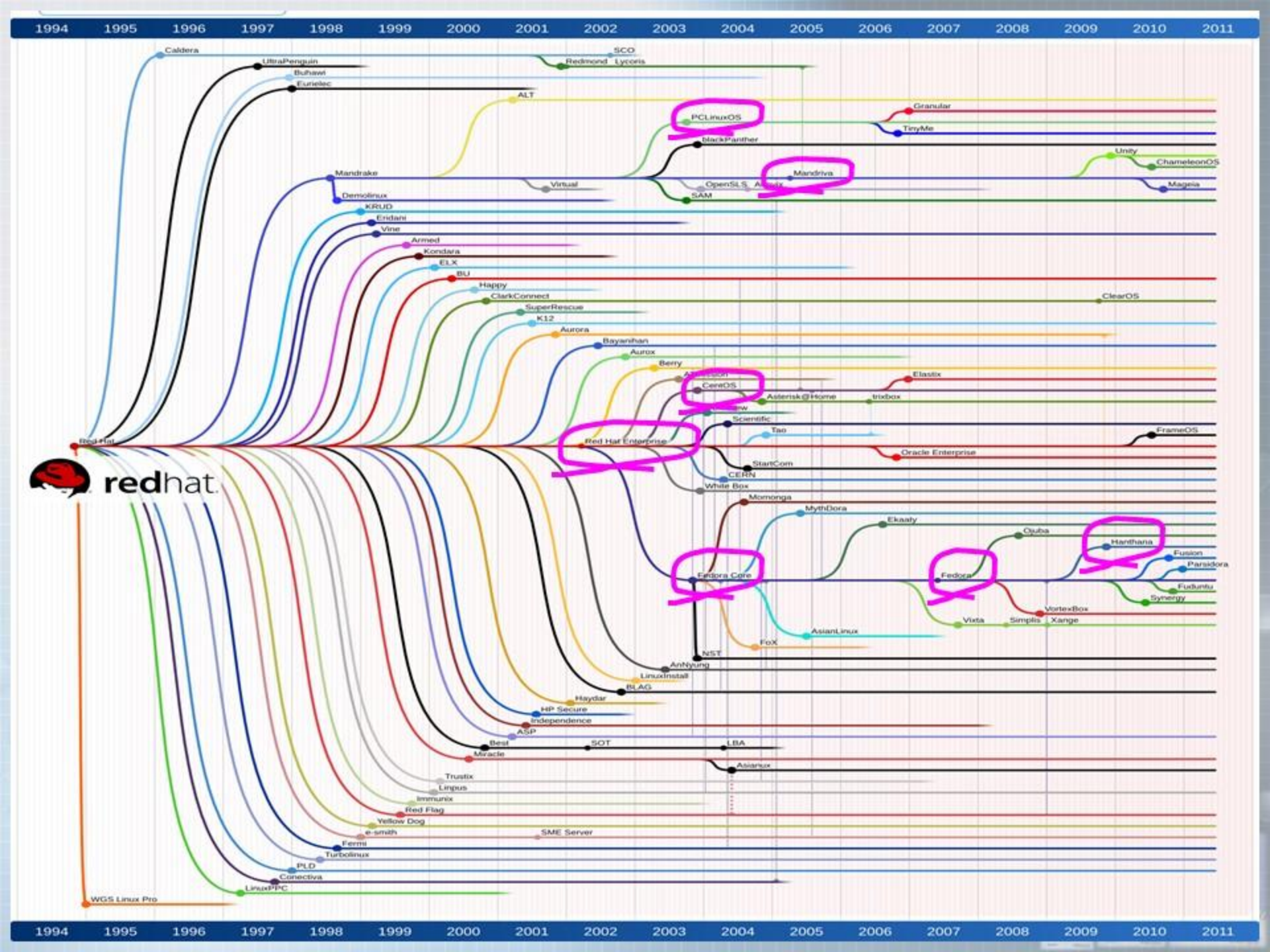


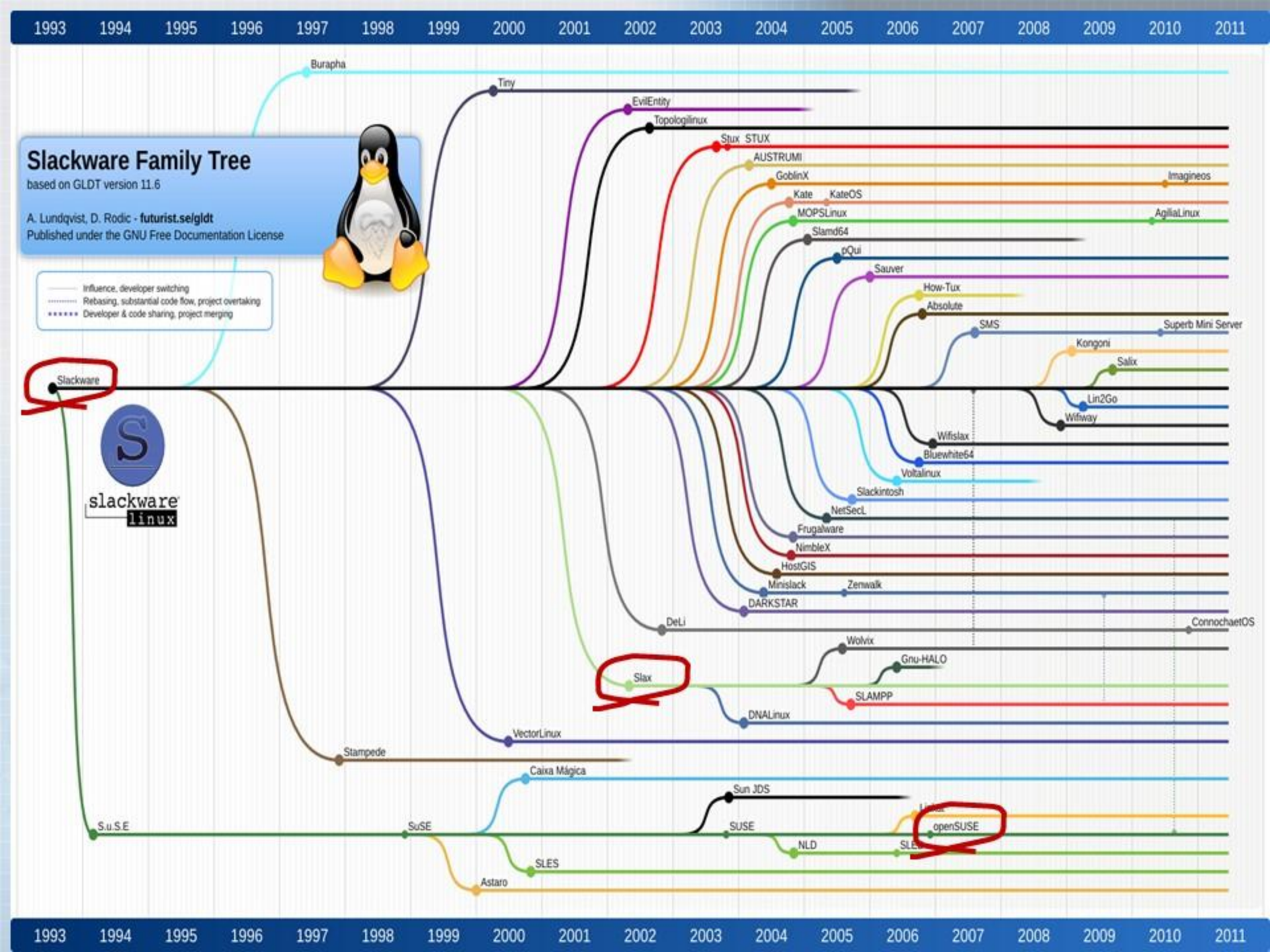


slackware linux




Linux distro timeline
Version 7.2 by NPU (nonplusx@gmail.com)
For the latest version, visit kde-files.org
Feel free to modify and spread. Mail me for updates, corrections and source file
Based on "Linea del tiempo Distribuciones Linux" by A. Sandoval (microteknologias.cl)
Additional info: distrowatch.com/wikipedia.org
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Edubuntu
Kubuntu
MEPIS
SimplyMEPIS
Damn Small Linux
Symphony OS
KNOPPIX
Kanotix
Morphix
LinEx
Debian
Progeny
Xandros
Corel
Yoper
Pardus
Puppy
Kate Linux
KateOS
Sorcerer
Source Mage
Lunar
TAMU
MCC Interim
SLS
Slackware
Vector
SLAX
Minislack
Zenwalk
Frugalware
S.u.S.E
SuSE
SUSE
openSUSE
Sun JDS
Jurix
Beehive
Enoch
Gentoo
RX4 / RX64
Sabayon
Kororaa
Stampede
Yggdrasil
VidaLinux
Arch
CRUX
Rock Linux
Linux From Scratch
dyne:bolic
Ark
LST
Caldera
SCO
Redmond
Lycoris
Conectiva
Mandrake
Mandriva
Virtual
PCLinuxOS
CentOS
United Linux
Scientific
White Box
Specifix
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SELinux
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OpenSLS
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Kondara
ELX
BU
Happy
ClarkConnect
SuperRescue
K12
Aurora
Bayanihan
Aurox
Berry
ClearOS
CereOS
Asterisk@Home
trixbox
Elastix
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Red Hat Enterprise
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LinuxInstall
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HP Secure
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LBA
Miracle
Asianux
Trustix
Linpus
Immunix
Red Flag
Yellow Dog
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SME Server
Fermi
Turbolinux
PLD
Conectiva
LinuxPPC
WGS Linux Pro


1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011
Slackware Family Tree
based on GLDT version 11.6
A. Lundqvist, D. Rodic - futurist.se/gldt
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Influence, developer switching
Rebasing, substantial code flow, project overtaking
Developer & code sharing, project merging
slackware linux
Slackware
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EvilEntity
Topologilinux
Stux
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GoblinX
Kate
KateOS
MOPSLinux
Slamd64
zQui
Sauver
How-Tux
Absolute
SMS
Superb Mini Server
Kongoni
Salix
Lin2Go
Wifiway
Wifislax
Bluewhite64
Voltalinux
Slackintosh
NetSecL
Frugalware
NimbleX
HostGIS
Minislack
Zenwalk
DARKSTAR
DeLi
ConnochaetOS
Wolvix
Gnu-HALO
Slax
SLAMPP
DNALinux
VectorLinux
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Caixa Mágica
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AgiliaLinux


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Debian Family Tree
based on GLDT version 11.6
A. Lundqvist, O. Rodic - futurist.se/gldt
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Debian
debian
Libranet
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Lindows
Linspire
Freespire
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SimplyMEPIS
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DeskEDU
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Ubuntu
Kubuntu
Edubuntu
Xubuntu
Mythbuntu
Christian Edition
Linux Mint
Element
Ubongo
Lycoris Edition
Lycoris Desktop/LX
Myddonia
Dreamlinux
OpenGEU
Foobuntu
Eeebuntu
Zeebuntu
Zevenos
Ultimate Edition
Mandriva
Aurora OS
Zenwalk
Slax OS
Vector
Wolvix
Vinux
Netrunner
Elive
Arch OS
ArchBang
Frugalware
Pingus OS
Absolut
Black Hat
Chromium OS
Lubuntu
Peppermint OS
Zenwalk
eBox
MoonOS
Ubuntu 666
Easy Peasy
Linexus
Ubuntu Rescue Remix
GuestRang
gOS
Kiwi
Ubuntulite
U Me
Maemo
Nokia
Maemo Edition
Nokia
gNewSense
PUD
Fluxbuntu
gPLAI
Arabian
Mokia
Blankon
Tregaid
Tuquito
Red Sleeves
Edubrisa
Pureos
Bliss
Mobian
MeeGo
GAI Pan Mobile
Inquisitor
Nova
DoudouLinux
Ubuntu Kylin
Cutelinux
Sabily
Progeny
Canonical
Metamorphose
Kanotix Hoa
Pelegrino
Webconverger
Librassoc
Mepis Lite
Nepalinux
Xandros
Elive
Mamma
Vinagre
NexAge
Evolva
grml
Caixadia Live
Eurielec
K-DeMar
Blueonyx
Aptosid
Debian Edu
Skolelinux
Progeny
DeMuDi
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Linpus
Guardian
Damn Small Linux
Damn Vulnerable Linux
Fedora
Kanotix
Auditor Security Linux
BackTrack
Backtrack
BSD
Knoppix
Whoppix
WHAX
Parallelknoppix
Morpix
Kaella
Aviena
Whiskey
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KnoppMyth
Aquamorph
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Kalango
Pocketlinux
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Linux Distributers

* Main Linux Distributers
  - Free & Community
  - Commercial

Linux Distributors

* **Free Community**
  - Fedora
  - Ubuntu
  - CentOS
  - Slackware
  - Linux Mint











* **Commercial**
  * Red Hat
  * SUSE
  * Oracle Linux







Interfaces

* Non-GUI
* GUI
* Graphical Environments

Interfaces



- Non-GUI interface
- GUI (Graphical)interface
- Graphical Environments
  - GNOME 1… 2… 3…
  - KDE
  - Cinnamon
  - Mate
  - fluxbox
  - xfce
  - zorin
  - enlightenment
  - lxde

Basement of Opensource OS

* Partition Structure
* Directory Structure
* Root Directory for Entire file system
* Files & Directory Names

Partition/Directory Structure

- A swap partition — Virtual memory: data is written to a swap partition when there is not enough RAM to store the data.

- A /boot/ partition (250 MB) — Contains the operating system kernel (which allows your system to boot OS).

- / partition



Partition/Directory Structure

Amount of RAM in the system	Recommended swap space	Recommended swap space if a llowing for hibernation
⩽ 2GB	2 times the amount of RAM	3 times the amount of RAM
> 2GB – 8GB	Equal to the amount of RAM	2 times the amount of RAM
> 8GB – 64GB	At least 4 GB	1.5 times the amount of RAM
> 64GB	At least 4 GB	Hibernation not recommended

Important — Supported file systems

The `/boot` and `/` (root) partition in Red Hat Enterprise Linux 6.7 can only use the ext2, ext3, and ext4 (recommended) file systems. You cannot use any other file system for this partition, such as Btrfs, XFS, or VFAT. Other partitions, such as `/home`, can use any supported file system, including Btrfs and XFS (if available). See the following article on the Red Hat Customer Portal for additional information:

Directory Structure

- Mount Point
- /
- Root Directory (/root)

Root User

Home Directory (~)



Directory Structure



root directory for the entire file system

bin/	Essential user command
boot/	Boot loader's files
dev/	Device files
etc/	System configuration files
home/	User home directories
lib/	Essential shared libraries and kernel module
root/	Super user's home directory
proc/	Virtual files/process status text files
media/	Mount point for removable media
mnt/	Mount point for the temporary file systems
opt/	Other application
sbin/	System binaries
srv/	Data for services
tmp/	Temporary files
usr/	Multiuser utilities and applications
var/	Variable files

Files and Directory Names

- Names may be up to 255 characters
- Can be use all characters except "/"
- Names are case sensitive
  - ex. MAIL, Mail, mAIL, mail



Command to Open Source OS



* All Command Case sensitive.
* First word accept as Command.
* Each item separated by space.
* Use options together or separate.
* We can chain command with;
* Use Option fully

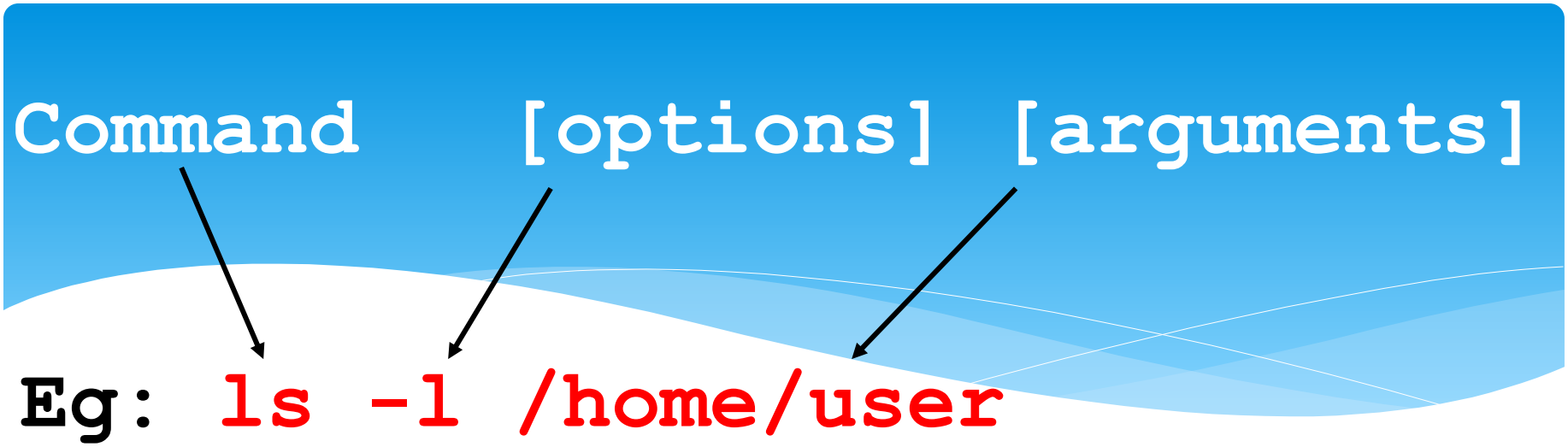
- **`command:`**

  This is the name of the executable program or built-in shell function you want to run.
  - Examples include **`ls`, `cd`, `mkdir`, `cp`, `rm`, `cat`, etc.**

- **`[options]:`**
  - Options, also known as flags or switches, modify the behavior of the command.
  - They typically begin with a hyphen (`-`) for short options or two hyphens (`--`) for long options.
  - **Short options**: Usually a single letter, often grouped together.
    - For example, `ls -lha` combines the `-l` (long listing format), `-h` (human-readable sizes), and `-a` (show all files, including hidden ones) options.
  - **Long options**: More descriptive words.
    - For example, `ls --all --human-readable --long` achieves the same as `ls -lha`.

- **`[arguments]:`**
  - Arguments are the inputs the command operates on.
  - These can be file paths, directory names, user names, or other data relevant to the command's function.
    - For example:

      in *`cp file1.txt /home/user/documents/`*, *`file1.txt`* and `/home/user/documents/` **are arguments representing the source** and **destination for the copy operation.**



- **Command (ls):** The core program or instruction telling the computer what action to take (list directory contents).
- **Option (-l)**: A modifier starting with a hyphen (-) or double-hyphen (--) that changes *how* the command behaves (displays long/detailed format).
- **Argument (/home/user)**: The specific data, file, or path that the command operates on

Thank You